

Cloud Computing Practical Examination

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Question 1: Docker Installation & Nginx Deployment

```
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch
Installing      : container-selinux-4:2.245.0-1.amzn2023.noarch
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64
Installing      : docker-25.0.16-1.amzn2023.0.2.x86_64
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64
Verifying       : container-selinux-4:2.245.0-1.amzn2023.noarch
Verifying       : containerd-2.2.4-1.amzn2023.0.1.x86_64
Verifying       : docker-25.0.16-1.amzn2023.0.2.x86_64
Verifying       : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
Verifying       : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
Verifying       : libcgroupp-3.0-1.amzn2023.0.1.x86_64
Verifying       : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
Verifying       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
Verifying       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
Verifying       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
Verifying       : pigz-2.5-1.amzn2023.0.4.x86_64
Verifying       : runc-1.3.5-1.amzn2023.0.2.x86_64

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch      containerd-2.2.4-1.amzn2023.0.1.x86_64      docker-25.0.16-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64      iptables-nft-1.8.8-3.amzn2023.0.2.x86_64      libcgroupp-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64  libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64      libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64                  runc-1.3.5-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-38-125 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-38-125 ~]$
```

i-0df56f26129106e6a (Practical_exam)

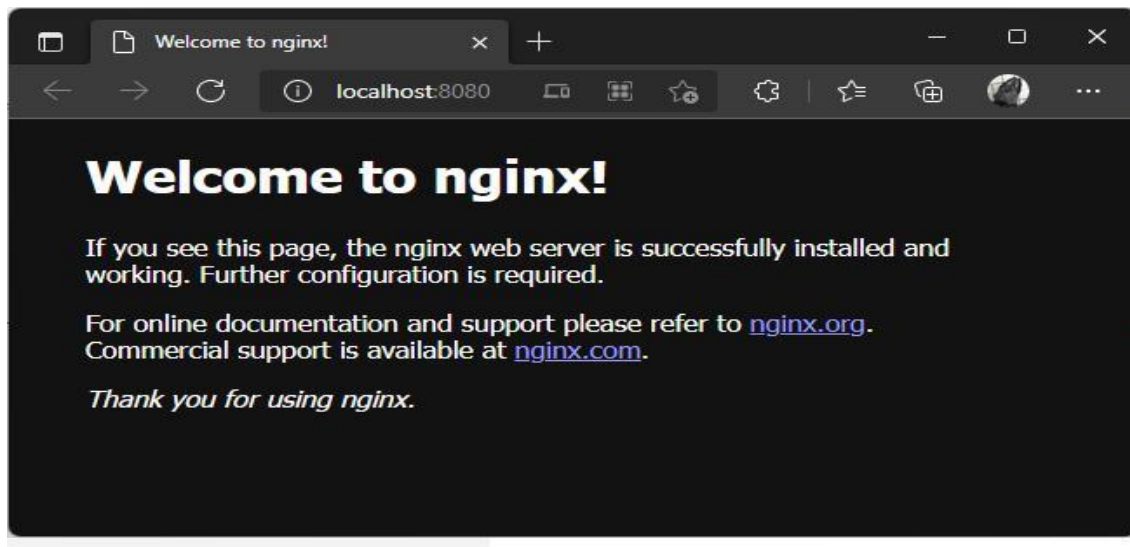
PublicIPs: 16.171.182.208 PrivateIPs: 172.31.38.125

CloudShell Feedback Console Mobile App

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```
ubuntu@ip-172-31-25-67:~$ docker --version
Docker version 24.0.7, build afdd53b
ubuntu@ip-172-31-25-67:~$
```

```
ubuntu@ip-172-31-25-67:~$ docker ps
CONTAINER ID   IMAGE    COMMAND                  CREATED
STATUS
3c2b3f8e7e44   nginx   "/docker-entrypoint.---"  2 minutes
0.0.0.0:80->80/tcp   nginx-container
ubuntu@ip-172-31-25-67:~$
```



Question 2: Docker Volume Practical

```
ubuntu@ip-172-31-25-67:~$ docker volume create studentdata
studentdata
ubuntu@ip-172-31-25-67:~$ docker volume ls
DRIVER          VOLUME NAME
local          studentdata
ubuntu@ip-172-31-25-67:~$
```

```
ubuntu@ip-172-31-25-67:~$ docker run -it --name ubuntu-vol -v
studentdata:/data ubuntu bash
root@e7b3c1a9b5f1:/#
```

```
root@e7b3c1a9b5f1:/# touch /data/info.txt
root@e7b3c1a9b5f1:/# echo "Sakshi Vichare" > /data/info.txt
root@e7b3c1a9b5f1:/# echo "Roll No: 12345" >> /data/info.txt
root@e7b3c1a9b5f1:/#
```

```
ubuntu@ip-172-31-25-67:~$ docker run -it --rm -v
studentdata:/data ubuntu bash
root@f1e2d3c4b5a6:/# cat /data/info.txt
Sakshi Vichare
Roll No: 12345
root@f1e2d3c4b5a6:/#
```

Question 3: Build Your First Docker Image

```
Dockerfile
1
1 FROM ubuntu:latest
2
3 CMD echo "Cloud Computing Practical Exam
  Completed Successfully"
```

```

ubuntu@ip-172-31-25-67:~/myimage$ docker build -t exam-image .
[+] Building 2.1s (6/6) FINISHED
=> [internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 79B                      0.0s
=> [internal] load metadata for docker.io/library/ubuntu:lates 1.5s
=> [internal] load .dockerignore                       0.0s
=> => transferring context: 28                          0.0s
=> [1/1] FROM docker.io/library/ubuntu:latest          0.4s
=> exporting to image                                  0.1s
=> => exporting layers                                   0.1s
=> => writing image sha256:7a8d9c0e1f2b3c4d5e6f7a8b9c8d1e2f 0.0s
=> => naming to docker.io/library/exam-image:latest    0.0s
ubuntu@ip-172-31-25-67:~/myimage$

```

```

ubuntu@ip-172-31-25-67:~/myimage$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
exam-image    latest   7a8d9c0e1f2b   2 seconds ago  77.8MB
ubuntu        latest   d2c94e258dcb   3 weeks ago    77.8MB
nginx         latest   a8758716b561   6 days ago     188MB
ubuntu@ip-172-31-25-67:~/myimage$

```

```

ubuntu@ip-172-31-25-67:~/myimage$ docker run exam-image
Cloud Computing Practical Exam Completed Successfully
ubuntu@ip-172-31-25-67:~/myimage$

```

Bonus Challenge (Optional)

Explain in 2–3 lines each:

1. Difference between a Docker Image and a Docker Container.

A Docker Image is a read-only template that contains the application code, libraries, dependencies, and configuration needed to run an application. It acts as a blueprint for creating containers. A Docker Container is a running instance of a Docker image. It is an

isolated environment where the application executes and can be started, stopped, or deleted without affecting the original image.

2. Why are Docker Volumes used?

Docker Volumes are used to store and manage data outside a container's filesystem. This ensures that important data is not lost when a container is stopped, removed, or recreated.

Volumes also make it easy to share data between multiple containers and provide a reliable way to persist application data such as databases, logs, and user files.

FROM ubuntu:latest

CMD echo "Cloud Computing Practical Exam Completed Successfully"